

API Reference

Product: TracSOPro

Base URL: <https://api.tracsopro.com/api> (production) or <http://localhost:3000/api> (local)

Last updated: June 2026

Overview

The TracSOPro backend is a REST API built with Express and TypeScript. All routes are prefixed with `/api` unless noted otherwise.

Item	Value
Interactive docs	Swagger UI at <code>/api-docs</code> when the server is running
Auth scheme	<code>Authorization: Bearer <access_token></code>
Response shape	<code>{ success: boolean, data?: any, message?: string, error?: string }</code>
WebSocket	Socket.io on the same host as the API (see WebSocket events)

Authentication

Public endpoints (`/api/auth`)

Method	Path	Description
GET	<code>/maintenance-status</code>	Platform maintenance flag
POST	<code>/register</code>	Register guard, client, or admin
POST	<code>/login</code>	Returns access + refresh tokens
POST	<code>/refresh</code>	Rotate access token
POST	<code>/logout</code>	Revoke refresh token
POST	<code>/verify-otp</code>	Verify email OTP
POST	<code>/resend-otp</code>	Resend OTP email
POST	<code>/forgot-password</code>	Request password reset
POST	<code>/reset-password</code>	Reset password with token

Authenticated endpoints (`/api/auth`)

Method	Path	Description
GET	<code>/me</code>	Current user profile
PUT	<code>/profile</code>	Update profile

Method	Path	Description
POST	/change-password	Change password

Registration notes:

- **GUARD** and **CLIENT** require a valid **invitationCode** from their security company.
- **ADMIN** registration creates a new **SecurityCompany** and links the user as owner.

Users (/api/users)

Method	Path	Auth	Description
POST	/profile-picture	JWT	Upload profile picture (multipart)
PATCH	/profile-picture	JWT	Update profile picture
DELETE	/profile-picture	JWT	Remove profile picture

Guards (/api/guards)

Method	Path	Roles	Description
GET	/	ADMIN	List guards
GET	/:id	ADMIN	Guard details
PUT	/profile	GUARD	Update own profile
PUT	/:id	ADMIN	Update guard
DELETE	/:id	ADMIN	Deactivate/remove guard
POST	/:id/emergency-contacts	ADMIN, GUARD	Add emergency contact
POST	/:id/qualifications	ADMIN, GUARD	Add qualification
GET	/:id/performance	ADMIN, GUARD	Performance metrics

Clients (/api/clients)

Method	Path	Roles	Description
GET	/	ADMIN	List clients
GET	/stats	ADMIN	Client statistics
GET	/:id	ADMIN	Client details
GET	/my-profile	CLIENT	Own profile
PUT	/profile	CLIENT	Update own profile
GET	/dashboard/stats	CLIENT	Dashboard metrics
GET	/my-guards	CLIENT	Guards assigned to client sites

Method	Path	Roles	Description
GET	/guards/:guardId	CLIENT	Guard detail
GET	/my-reports	CLIENT	Incident/shift reports
GET	/my-sites	CLIENT	Client sites
GET	/my-shifts	CLIENT	Client shifts
GET	/my-notifications	CLIENT	Notifications
PUT	/reports/:reportId/respond	CLIENT	Respond to a report
POST	/shifts	CLIENT	Create shift
POST	/shifts/bulk	CLIENT	Bulk create shifts
PUT	/shifts/:shiftId	CLIENT	Update shift
DELETE	/shifts/:shiftId	CLIENT	Cancel shift
PUT	/:id	ADMIN	Update client
DELETE	/:id	ADMIN	Remove client

Sites (/api/sites)

Method	Path	Roles	Description
POST	/	CLIENT	Create site
GET	/my-sites	CLIENT	List own sites
GET	/active	GUARD, ADMIN	Active sites
GET	/:id	JWT	Site details
PUT	/:id	CLIENT	Update site
DELETE	/:id	CLIENT	Delete site

Shifts — guard (/api/shifts)

Method	Path	Description
GET	/active	Current active shift
GET	/upcoming	Upcoming shifts
GET	/today	Today's shifts
GET	/past	Past shifts
GET	/weekly-summary	Weekly summary
GET	/schedule/30-days	30-day schedule

Method	Path	Description
GET	/statistics	Guard shift stats
GET	/:id	Shift details
GET	/:id/active-break	Active break on shift
POST	/:id/check-in	Check in (GPS validated)
POST	/:id/check-out	Check out
POST	/:id/start-break	Start break
POST	/:shiftId/end-break/:breakId	End break
POST	/:id/report-incident	Report shift incident

Shifts — admin ([/api/admin/shifts](#))

Method	Path	Roles	Description
GET	/	ADMIN	List shifts
GET	/schedule/30-days	ADMIN	30-day schedule
GET	/unassigned	ADMIN	Unassigned shifts
POST	/	ADMIN	Create shift
POST	/bulk	ADMIN	Bulk create
PATCH	/:shiftId/assign-guard	ADMIN	Assign guard
PUT	/:shiftId	ADMIN	Update shift
DELETE	/:shiftId	ADMIN	Delete shift

Shift reports ([/api/shift-reports](#))

Method	Path	Roles	Description
GET	/company	ADMIN	Company shift reports
POST	/	GUARD	Submit report
GET	/	GUARD	List own reports
GET	/:id	GUARD	Report detail
PUT	/:id	GUARD	Update report
DELETE	/:id	GUARD	Delete report
GET	/shift/:shiftId	JWT	Reports for a shift

Tracking ([/api/tracking](#))

Method	Path	Roles	Description
POST	/location	GUARD	Upload GPS point
POST	/geofence-event	GUARD	Record geofence enter/exit
POST	/check-geofences/:guardId	GUARD	Evaluate geofences
GET	/history/:guardId	ADMIN, CLIENT	Location history
GET	/:guardId/latest	ADMIN, CLIENT	Latest location
GET	/live-locations	ADMIN, CLIENT	All live guard locations
GET	/geofence-events/:guardId	ADMIN, CLIENT	Geofence events
GET	/real-time-data	ADMIN, CLIENT	Real-time tracking bundle
GET	/analytics	ADMIN	Tracking analytics

Incidents (/api/incidents)

Method	Path	Roles	Description
GET	/	ADMIN, SUPER_ADMIN	List incidents
GET	/stats	ADMIN, SUPER_ADMIN	Incident statistics
GET	/:id	JWT	Incident detail
POST	/	JWT	Create incident
PUT	/:id	ADMIN, SUPER_ADMIN	Update incident
POST	/:id/evidence	JWT	Upload evidence

Incident reports (/api/incident-reports)

Method	Path	Roles	Description
POST	/	GUARD	Submit incident report
GET	/	GUARD	List own reports
GET	/:id	JWT	Report detail
PUT	/:id	GUARD	Update report
DELETE	/:id	GUARD	Delete report
PUT	/:id/respond	ADMIN, CLIENT	Respond to report
GET	/admin/all	ADMIN	All company reports
GET	/admin/stats	ADMIN	Report statistics

Emergency (/api/emergency)

Method	Path	Roles	Description
POST	/alert	GUARD	Trigger emergency alert
GET	/my-active	GUARD	Active alert for guard
POST	/alert/:alertId/acknowledge	ADMIN, CLIENT	Acknowledge alert
POST	/alert/:alertId/resolve	ADMIN	Resolve alert
GET	/alerts/active	ADMIN, CLIENT	Active alerts
GET	/guard/:guardId/history	Mixed	Alert history
GET	/statistics	Mixed	Emergency statistics

Notifications (/api/notifications)

Method	Path	Description
GET	/	List notifications
GET	/stats	Unread counts
GET	/push-status	FCM registration status
POST	/test-push	Send test push
POST	/register-device	Register FCM device token
POST	/record-event	Record notification event
PUT	/:id/read	Mark as read
PUT	/read-all	Mark all as read
DELETE	/	Clear all
DELETE	/:id	Delete notification

Chat (/api/chat)

Method	Path	Description
GET	/	List conversations
GET	/search	Search conversations
GET	/support	Support conversation
POST	/	Create conversation
POST	/support	Open support chat
POST	/support/company	Company support chat
GET	/:chatId	Conversation detail

Method	Path	Description
GET	/:chatId/messages	Message history
POST	/:chatId/messages	Send message
POST	/:chatId/read	Mark messages read

Settings ([/api/settings](#))

Method	Path	Roles	Description
GET	/notifications	JWT	Notification preferences
PUT	/notifications	JWT	Update preferences
GET	/profile	JWT	Settings profile
PUT	/profile	JWT	Update settings profile
POST	/support/contact	JWT	Contact support
GET	/support/tickets	JWT	List support tickets
GET	/support/tickets/:id	JWT	Ticket detail
GET	/attendance-history	GUARD	Attendance history
GET	/past-jobs	GUARD	Past jobs
GET	/company	CLIENT	Company settings
PUT	/company	CLIENT	Update company settings
POST	/change-password	JWT	Change password

Payments ([/api/payments](#))

Method	Path	Roles	Description
POST	/intent	CLIENT	Create payment intent
POST	/setup-intent	CLIENT	Setup payment method
POST	/auto-pay	CLIENT	Enable auto-pay
POST	/invoice	ADMIN	Create invoice
POST	/invoice/monthly	ADMIN	Generate monthly invoice
GET	/methods	CLIENT, ADMIN	Payment methods
GET	/invoices	CLIENT, ADMIN	Invoice list
GET	/plans	ADMIN, SUPER_ADMIN	Subscription plans
POST	/subscriptions/checkout	ADMIN, SUPER_ADMIN	Stripe checkout

Method	Path	Roles	Description
POST	/portal	ADMIN, SUPER_ADMIN	Billing portal session

Stripe webhook (not under [/api](#) prefix): [POST /api/payments/webhook](#) — registered on the Express app root with raw body parsing for Stripe signature verification.

Admin ([/api/admin](#))

Method	Path	Description
GET	/subscription	Company subscription
GET	/dashboard/stats	Dashboard statistics
GET	/dashboard/activity	Recent activity
GET	/company	Company profile
PUT	/company	Update company

Admin users ([/api/admin/users](#))

CRUD for company users, plus [PATCH /:id/status](#) for activate/deactivate.

Admin sites ([/api/admin/sites](#))

CRUD for company sites.

Admin clients ([/api/admin/clients](#))

List and manage client relationships.

Admin operations ([/api/admin/operations](#))

Method	Path	Description
GET	/metrics	Operations metrics
GET	/guards	Guards on duty
GET	/activity	Live activity feed

Admin invitations ([/api/admin/invitations](#))

Create, list, revoke invitation codes for guard/client onboarding.

Super Admin ([/api/super-admin](#))

All routes require [SUPER_ADMIN](#) role.

Area	Key endpoints
Overview	GET /overview

Area	Key endpoints
Companies	GET/POST /companies, GET/PUT/PATCH /companies/:id, subscription & checkout
Analytics	GET /analytics
Billing	GET /billing, /payments, /payments/analytics, /:id, status patch
Platform	GET/PUT /settings, POST /export-data, GET /audit-logs
Users	GET /users, POST /impersonate

Health and legacy

Method	Path	Description
GET	/health	Server health check
GET	/locations	Legacy stub (returns [])
GET/POST	/messages	Legacy stub

WebSocket (Socket.io)

Connect to the same host as the API (e.g. <https://api.tracsopro.com>). After connecting, emit `authenticate` with `{ token, userId?, role? }`.

Client → Server

Event	Payload	Description
<code>authenticate</code>	<code>{ token, userId?, role? }</code>	JWT validation
<code>location_update</code>	GPS coordinates	Guard location upload
<code>geofence_event</code>	Geofence data	Enter/exit event
<code>emergency_alert</code>	Alert payload	Emergency broadcast
<code>shift_status_update</code>	Shift status	Shift lifecycle update
<code>request_live_locations</code>	—	Request current locations
<code>join_room</code>	<code>{ roomId }</code>	Join chat or ops room
<code>leave_room</code>	<code>{ roomId }</code>	Leave room

Server → Client

Event	Description
<code>authenticated / authentication_error</code>	Auth result
<code>guard_location_update</code>	Single guard location (admins)
<code>live_locations_update</code>	Periodic broadcast (every ~30s)

Event	Description
<code>live_locations_data</code>	Response to <code>request_live_locations</code>
<code>geofence_event</code>	Geofence notification
<code>emergency_alert</code> / <code>emergency_broadcast</code>	Emergency notifications
<code>shift_status_changed</code>	Shift status change
<code>room_joined</code> / <code>room_left</code>	Room membership
<code>notification</code>	In-app notification push

Error codes

HTTP	Meaning
400	Validation error
401	Missing or invalid token
403	Insufficient role or tenant access
404	Resource not found
429	Rate limit exceeded
503	Maintenance mode (non-super-admin blocked)

Related documents

- [SYSTEM_ARCHITECTURE.md](#) — system overview
- [DATABASE_SCHEMA.md](#) — data models
- Swagger UI at `/api-docs` — full request/response schemas